AIDCF

AI Data Centre Facilities Assurance Framework

END-TO-END TRACEABILITY & MACHINE-EXECUTABLE DATA STANDARD

Version 1.0

A canonical information contract for identifiers, lineage, schemas, graphs, validation, exchange, change impact and conformance

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Document control

Normative requirements.

Element	Requirement
Purpose	Define canonical objects, relationships, IDs, traceability, serialization, validation, security and exchange.
Outputs	Registries, JSON Schema, JSON-LD, RDF/OWL, SHACL, mappings, events, tests and fixtures.
Authority	Tools implement declared rules; accountable people approve material facts and decisions.
Conflict	Contradictory sources create BaselineIssue records.
Boundary	Technical conformance is not engineering approval, assurance or certification.

Executive thesis

Every material conclusion must be reconstructable from stable identities, approved source versions, explicit relationships, evidence and review. Missing or conflicting data remains UNKNOWN or a validation failure.

Canonical chain

Capability → Outcome → Function → Architecture → Dependency/Fault Domain → Failure → Safeguard → Control → Evidence Requirement → Evidence → Assessment → Result → Finding/UNKNOWN → Gate → Score → Recommendation → Decision → Report → Trigger

1 Authority and precedence

Normative requirements.

Element	Requirement
Semantic owner retained	Required
Schemas cannot weaken doctrine	Required
No silent reconciliation	Required
Release manifest pins versions and hashes	Required
Technology neutral	Required
Human approval	Required
History preserved	Required
Validity is not assurance	Required

2 Constitutional invariants

Normative requirements.

Element	Requirement
Capability/outcomes primary	Required
Assertions need evidence/review	Required
UNKNOWN first-class	Required
Gates separate from scores	Required
Assurance dimensions distinct	Required
Dependencies visible	Required
Lineage mandatory	Required
Change can invalidate	Required
Automation proposes	Required
AIDCF/AITG conclusions separate	Required
Connectivity is not causality	Required
Sensitive paths controlled	Required

3 Conformance levels

Normative requirements.

Element	Requirement
T0 vocabulary	Required
T1 records and IDs	Required
T2 traceability	Required
T3 JSON validation	Required
T4 graph validation	Required
T5 decision-ready	Required
T6 continuing assurance	Required
T7 test-suite conformance	Required

4 Release manifest

Normative requirements.

Element	Requirement
releaseId	Required
artifactManifest	Required
registryVersions	Required
schemaVersions	Required
approved cardinality	Required
knownIssues	Required
migrationSet	Required
testSuite	Required
approvals	Required
integrity signature	Required

5 Identifier standard

Normative requirements.

Element	Requirement
Stable namespace or IRI	Required
Unique identity	Required
Meaning change creates new ID	Required
Version separate from identity	Required
Controlled codes	Required
Typed external IDs	Required
Reviewed aliases/shared identity	Required
Candidate IDs non-authoritative	Required
Collision blocker	Required

6 Baseline harmonisation

Normative requirements.

Element	Requirement
BaselineIssue	First-class conflict object
Trigger	Conflicting counts, IDs, titles, definitions, mappings or statuses
Status	BLOCKED or CONDITIONALLY_BASELINED
Resolution	Approved decision, registry update, migration and supersession
Preservation	Original values and source lineage retained

7 Canonical object registry

Normative requirements.

Element	Requirement
ReleaseManifest	Canonical class with semantic owner
BaselineIssue	Canonical class with semantic owner
CapabilityInstance	Canonical class with semantic owner
ServiceOutcome	Canonical class with semantic owner
RequiredFunction	Canonical class with semantic owner
ArchitectureSnapshot	Canonical class with semantic owner
InfrastructureSystem	Canonical class with semantic owner
AssetInstance	Canonical class with semantic owner
Interface	Canonical class with semantic owner
Resource	Canonical class with semantic owner
Dependency	Canonical class with semantic owner
FaultDomain	Canonical class with semantic owner
OperatingState	Canonical class with semantic owner
CapacityAssertion	Canonical class with semantic owner
Threat	Canonical class with semantic owner
Hazard	Canonical class with semantic owner
FailureMode	Canonical class with semantic owner
FailureScenario	Canonical class with semantic owner
Safeguard	Canonical class with semantic owner
ControlDomain	Canonical class with semantic owner
ControlFamily	Canonical class with semantic owner
Control	Canonical class with semantic owner
EvidenceRequirement	Canonical class with semantic owner
EvidenceObject	Canonical class with semantic owner
EvidencePackage	Canonical class with semantic owner
AssuranceProgram	Canonical class with semantic owner
Assessment	Canonical class with semantic owner
AssessmentProcedure	Canonical class with semantic owner
Sample	Canonical class with semantic owner
TestRecord	Canonical class with semantic owner
ControlAssessment	Canonical class with semantic owner
Finding	Canonical class with semantic owner
UnknownRecord	Canonical class with semantic owner
CriticalGate	Canonical class with semantic owner
GateResult	Canonical class with semantic owner
ScoreSnapshot	Canonical class with semantic owner
DecisionRecommendation	Canonical class with semantic owner
ReviewRecord	Canonical class with semantic owner
AssuranceDecision	Canonical class with semantic owner
AssuranceStatement	Canonical class with semantic owner
AssuranceReport	Canonical class with semantic owner
Action	Canonical class with semantic owner
ReassessmentTrigger	Canonical class with semantic owner
Event	Canonical class with semantic owner
IntegrationHandoff	Canonical class with semantic owner
ValidationResult	Canonical class with semantic owner
MigrationRecord	Canonical class with semantic owner
DistributionControl	Canonical class with semantic owner

8 Common object envelope

Normative requirements.

Element	Requirement
id	Mandatory unless explicit N/A
type	Mandatory unless explicit N/A
semanticOwner	Mandatory unless explicit N/A
version	Mandatory unless explicit N/A
status	Mandatory unless explicit N/A
sourceArtifact	Mandatory unless explicit N/A
sourceRecord	Mandatory unless explicit N/A
name	Mandatory unless explicit N/A
description	Mandatory unless explicit N/A
scope	Mandatory unless explicit N/A
effectiveFrom	Mandatory unless explicit N/A
effectiveTo	Mandatory unless explicit N/A
createdAt	Mandatory unless explicit N/A
createdBy	Mandatory unless explicit N/A
modifiedAt	Mandatory unless explicit N/A
modifiedBy	Mandatory unless explicit N/A
classification	Mandatory unless explicit N/A
provenance	Mandatory unless explicit N/A
reviewState	Mandatory unless explicit N/A
supersedes	Mandatory unless explicit N/A
externalIdentifiers	Mandatory unless explicit N/A
extensions	Mandatory unless explicit N/A

Object profile: Capability

Normative requirements.

Element	Requirement
owner	Required or explicit N/A/UNKNOWN
lifecycle	Required or explicit N/A/UNKNOWN
workload	Required or explicit N/A/UNKNOWN
architectureBaseline	Required or explicit N/A/UNKNOWN
outcomeIds	Required or explicit N/A/UNKNOWN
criticality	Required or explicit N/A/UNKNOWN
boundary	Required or explicit N/A/UNKNOWN
demandProfile	Required or explicit N/A/UNKNOWN
operatingEnvelope	Required or explicit N/A/UNKNOWN

Object profile: Architecture

Normative requirements.

Element	Requirement
capabilityId	Required or explicit N/A/UNKNOWN
systems	Required or explicit N/A/UNKNOWN
assets	Required or explicit N/A/UNKNOWN
interfaces	Required or explicit N/A/UNKNOWN
resources	Required or explicit N/A/UNKNOWN
dependencies	Required or explicit N/A/UNKNOWN
faultDomains	Required or explicit N/A/UNKNOWN
states	Required or explicit N/A/UNKNOWN
assertions	Required or explicit N/A/UNKNOWN
validation	Required or explicit N/A/UNKNOWN

Object profile: Dependency

Normative requirements.

Element	Requirement
sourceId	Required or explicit N/A/UNKNOWN
targetId	Required or explicit N/A/UNKNOWN
dependencyType	Required or explicit N/A/UNKNOWN
direction	Required or explicit N/A/UNKNOWN
conditions	Required or explicit N/A/UNKNOWN
criticality	Required or explicit N/A/UNKNOWN
substitutability	Required or explicit N/A/UNKNOWN
sharedMode	Required or explicit N/A/UNKNOWN
evidenceRefs	Required or explicit N/A/UNKNOWN
confidence	Required or explicit N/A/UNKNOWN

Object profile: FailureScenario

Normative requirements.

Element	Requirement
preState	Required or explicit N/A/UNKNOWN
initiatingSource	Required or explicit N/A/UNKNOWN
steps	Required or explicit N/A/UNKNOWN
safeguards	Required or explicit N/A/UNKNOWN
residualState	Required or explicit N/A/UNKNOWN
outcomeEffects	Required or explicit N/A/UNKNOWN
evidenceRefs	Required or explicit N/A/UNKNOWN
confidence	Required or explicit N/A/UNKNOWN

Object profile: Safeguard

Normative requirements.

Element	Requirement
function	Required or explicit N/A/UNKNOWN
protectedSubject	Required or explicit N/A/UNKNOWN
activation	Required or explicit N/A/UNKNOWN
authority	Required or explicit N/A/UNKNOWN
dependencies	Required or explicit N/A/UNKNOWN
stateCoverage	Required or explicit N/A/UNKNOWN
failureModes	Required or explicit N/A/UNKNOWN
evidenceRefs	Required or explicit N/A/UNKNOWN
controlRefs	Required or explicit N/A/UNKNOWN

Object profile: Control

Normative requirements.

Element	Requirement
domainId	Required or explicit N/A/UNKNOWN
familyId	Required or explicit N/A/UNKNOWN
objective	Required or explicit N/A/UNKNOWN
requirement	Required or explicit N/A/UNKNOWN
applicability	Required or explicit N/A/UNKNOWN
criteria	Required or explicit N/A/UNKNOWN
methods	Required or explicit N/A/UNKNOWN
assuranceDimensions	Required or explicit N/A/UNKNOWN
evidenceProfile	Required or explicit N/A/UNKNOWN
gateRefs	Required or explicit N/A/UNKNOWN

Object profile: Evidence

Normative requirements.

Element	Requirement
evidenceClass	Required or explicit N/A/UNKNOWN
source	Required or explicit N/A/UNKNOWN
collector	Required or explicit N/A/UNKNOWN
custodian	Required or explicit N/A/UNKNOWN
assertionRefs	Required or explicit N/A/UNKNOWN
scope	Required or explicit N/A/UNKNOWN
period	Required or explicit N/A/UNKNOWN
state	Required or explicit N/A/UNKNOWN
integrity	Required or explicit N/A/UNKNOWN
qualityVector	Required or explicit N/A/UNKNOWN
sufficiency	Required or explicit N/A/UNKNOWN
restrictions	Required or explicit N/A/UNKNOWN

Object profile: Assessment

Normative requirements.

Element	Requirement
programId	Required or explicit N/A/UNKNOWN
decisionQuestion	Required or explicit N/A/UNKNOWN
capabilityId	Required or explicit N/A/UNKNOWN
scope	Required or explicit N/A/UNKNOWN
profile	Required or explicit N/A/UNKNOWN
methods	Required or explicit N/A/UNKNOWN
team	Required or explicit N/A/UNKNOWN
evidenceManifest	Required or explicit N/A/UNKNOWN
results	Required or explicit N/A/UNKNOWN
gates	Required or explicit N/A/UNKNOWN
review	Required or explicit N/A/UNKNOWN

Object profile: Finding

Normative requirements.

Element	Requirement
criterionRef	Required or explicit N/A/UNKNOWN
condition	Required or explicit N/A/UNKNOWN
evidenceRefs	Required or explicit N/A/UNKNOWN
cause	Required or explicit N/A/UNKNOWN
consequence	Required or explicit N/A/UNKNOWN
classification	Required or explicit N/A/UNKNOWN
gateEffect	Required or explicit N/A/UNKNOWN
scoreEffect	Required or explicit N/A/UNKNOWN
owner	Required or explicit N/A/UNKNOWN
status	Required or explicit N/A/UNKNOWN

Object profile: UNKNOWN

Normative requirements.

Element	Requirement
subjectRef	Required or explicit N/A/UNKNOWN
missingAttribute	Required or explicit N/A/UNKNOWN
reason	Required or explicit N/A/UNKNOWN
materiality	Required or explicit N/A/UNKNOWN
decisionEffect	Required or explicit N/A/UNKNOWN
evidenceNeeded	Required or explicit N/A/UNKNOWN
owner	Required or explicit N/A/UNKNOWN
trigger	Required or explicit N/A/UNKNOWN
status	Required or explicit N/A/UNKNOWN

Object profile: GateResult

Normative requirements.

Element	Requirement
gateId	Required or explicit N/A/UNKNOWN
applicability	Required or explicit N/A/UNKNOWN
criterion	Required or explicit N/A/UNKNOWN
evidenceRefs	Required or explicit N/A/UNKNOWN
stateCoverage	Required or explicit N/A/UNKNOWN
findings	Required or explicit N/A/UNKNOWN
unknowns	Required or explicit N/A/UNKNOWN
status	Required or explicit N/A/UNKNOWN
decisionEffect	Required or explicit N/A/UNKNOWN
review	Required or explicit N/A/UNKNOWN

Object profile: Score

Normative requirements.

Element	Requirement
subjectRef	Required or explicit N/A/UNKNOWN
profileVersion	Required or explicit N/A/UNKNOWN
inputs	Required or explicit N/A/UNKNOWN
weights	Required or explicit N/A/UNKNOWN
rawIndicator	Required or explicit N/A/UNKNOWN
adjustedIndicator	Required or explicit N/A/UNKNOWN
coverage	Required or explicit N/A/UNKNOWN
confidence	Required or explicit N/A/UNKNOWN
gateOverlay	Required or explicit N/A/UNKNOWN
unknownOverlay	Required or explicit N/A/UNKNOWN

Object profile: Decision

Normative requirements.

Element	Requirement
question	Required or explicit N/A/UNKNOWN
subjectRef	Required or explicit N/A/UNKNOWN
scope	Required or explicit N/A/UNKNOWN
recommendationRef	Required or explicit N/A/UNKNOWN
gateResults	Required or explicit N/A/UNKNOWN
unknowns	Required or explicit N/A/UNKNOWN
scores	Required or explicit N/A/UNKNOWN
conditions	Required or explicit N/A/UNKNOWN
decisionState	Required or explicit N/A/UNKNOWN
authority	Required or explicit N/A/UNKNOWN
validity	Required or explicit N/A/UNKNOWN

Object profile: Report

Normative requirements.

Element	Requirement
reportClass	Required or explicit N/A/UNKNOWN
programRef	Required or explicit N/A/UNKNOWN
assessmentRef	Required or explicit N/A/UNKNOWN
decisionRef	Required or explicit N/A/UNKNOWN
issuer	Required or explicit N/A/UNKNOWN
users	Required or explicit N/A/UNKNOWN
intendedUse	Required or explicit N/A/UNKNOWN
classification	Required or explicit N/A/UNKNOWN
validity	Required or explicit N/A/UNKNOWN
sections	Required or explicit N/A/UNKNOWN
approvals	Required or explicit N/A/UNKNOWN

Object profile: Event

Normative requirements.

Element	Requirement
eventType	Required or explicit N/A/UNKNOWN
subjectRef	Required or explicit N/A/UNKNOWN
source	Required or explicit N/A/UNKNOWN
observedAt	Required or explicit N/A/UNKNOWN
effectiveAt	Required or explicit N/A/UNKNOWN
payload	Required or explicit N/A/UNKNOWN
severity	Required or explicit N/A/UNKNOWN
triggerRule	Required or explicit N/A/UNKNOWN
reviewState	Required or explicit N/A/UNKNOWN
affectedAssertions	Required or explicit N/A/UNKNOWN

9 Relationship registry

Normative requirements.

Element	Requirement
HAS_OUTCOME	Typed, directed, scoped, versioned and reviewable
REQUIRES_FUNCTION	Typed, directed, scoped, versioned and reviewable
IMPLEMENTED_BY	Typed, directed, scoped, versioned and reviewable
COMPOSED_OF	Typed, directed, scoped, versioned and reviewable
CONNECTS_TO	Typed, directed, scoped, versioned and reviewable
CONSUMES_RESOURCE	Typed, directed, scoped, versioned and reviewable
DEPENDS_ON	Typed, directed, scoped, versioned and reviewable
SHARES_FAULT_DOMAIN_WITH	Typed, directed, scoped, versioned and reviewable
LOCATED_IN	Typed, directed, scoped, versioned and reviewable
OPERATED_BY	Typed, directed, scoped, versioned and reviewable
CONTROLLED_BY	Typed, directed, scoped, versioned and reviewable
MONITORED_BY	Typed, directed, scoped, versioned and reviewable
PROTECTED_BY	Typed, directed, scoped, versioned and reviewable
EXPOSED_TO	Typed, directed, scoped, versioned and reviewable
INITIATES	Typed, directed, scoped, versioned and reviewable
PROPAGATES_TO	Typed, directed, scoped, versioned and reviewable
RECOVERS_THROUGH	Typed, directed, scoped, versioned and reviewable
GOVERNED_BY	Typed, directed, scoped, versioned and reviewable
HAS_EVIDENCE_REQUIREMENT	Typed, directed, scoped, versioned and reviewable
EVIDENCED_BY	Typed, directed, scoped, versioned and reviewable
SUPPORTS	Typed, directed, scoped, versioned and reviewable
CHALLENGES	Typed, directed, scoped, versioned and reviewable
QUALIFIES	Typed, directed, scoped, versioned and reviewable
ASSESSED_BY	Typed, directed, scoped, versioned and reviewable
RESULTS_IN	Typed, directed, scoped, versioned and reviewable
HAS_FINDING	Typed, directed, scoped, versioned and reviewable
HAS_UNKNOWN	Typed, directed, scoped, versioned and reviewable
EVALUATES_GATE	Typed, directed, scoped, versioned and reviewable
CONSTRAINS	Typed, directed, scoped, versioned and reviewable
SUMMARISED_BY	Typed, directed, scoped, versioned and reviewable
RECOMMENDS	Typed, directed, scoped, versioned and reviewable
REVIEWED_BY	Typed, directed, scoped, versioned and reviewable
APPROVED_BY	Typed, directed, scoped, versioned and reviewable
REPORTED_IN	Typed, directed, scoped, versioned and reviewable
REQUIRES_ACTION	Typed, directed, scoped, versioned and reviewable
CLOSED_BY	Typed, directed, scoped, versioned and reviewable
INVALIDATED_BY	Typed, directed, scoped, versioned and reviewable
TRIGGERS_REASSESSMENT	Typed, directed, scoped, versioned and reviewable
SUPERSEDES	Typed, directed, scoped, versioned and reviewable
MIGRATED_FROM	Typed, directed, scoped, versioned and reviewable
INTEGRATION_HANDOFF	Typed, directed, scoped, versioned and reviewable

10 Relationship envelope

Normative requirements.

Element	Requirement
relationshipId	Required
type	Required
sourceId	Required
targetId	Required
semanticOwner	Required
scope	Required
conditions	Required
effectiveFrom	Required
effectiveTo	Required
sourceRecord	Required
evidenceRefs	Required
confidence	Required
reviewState	Required
classification	Required
version	Required
supersedes	Required

11 End-to-end traceability

Normative requirements.

Element	Requirement
TR-01	Capability to Outcome
TR-02	Outcome to Function
TR-03	Function to Architecture
TR-04	Architecture to Dependency
TR-05	Dependency to Failure
TR-06	Failure to Safeguard
TR-07	Safeguard to Control
TR-08	Control to Evidence Requirement
TR-09	Requirement to Evidence or UNKNOWN
TR-10	Procedure to Result
TR-11	Result to Finding or UNKNOWN
TR-12	Gate to Evidence/Findings/UNKNOWN
TR-13	Score to immutable inputs/profile
TR-14	Recommendation to constraints
TR-15	Decision to authority/scope/validity
TR-16	Report to approved decision
TR-17	Trigger to impacted objects
TR-18	Supersession to prior/successor

Controlled vocabulary: Lifecycle

Normative requirements.

Element	Requirement
DRAFT	Controlled code
CANDIDATE	Controlled code
REVIEWED	Controlled code
APPROVED	Controlled code
ACTIVE	Controlled code
SUSPENDED	Controlled code
WITHDRAWN	Controlled code
SUPERSEDED	Controlled code
RETIRED	Controlled code

Controlled vocabulary: Assertion

Normative requirements.

Element	Requirement
CANDIDATE	Controlled code
SUPPORTED	Controlled code
CHALLENGED	Controlled code
QUALIFIED	Controlled code
CONFLICTING	Controlled code
UNKNOWN	Controlled code
SUPERSEDED	Controlled code

Controlled vocabulary: ControlResult

Normative requirements.

Element	Requirement
NOT_ASSESSED	Controlled code
UNKNOWN	Controlled code
N/A	Controlled code
NOT_IMPLEMENTED	Controlled code
PARTIAL	Controlled code
IMPLEMENTED	Controlled code
OPERATING_EFFECTIVELY	Controlled code
INEFFECTIVE	Controlled code
SUSPENDED	Controlled code

Controlled vocabulary: GateResult

Normative requirements.

Element	Requirement
PASS	Controlled code
FAIL	Controlled code
UNKNOWN	Controlled code
APPROVED_N/A	Controlled code
NOT_ASSESSED	Controlled code

Controlled vocabulary: Decision

Normative requirements.

Element	Requirement
REASONABLY_ASSURED	Controlled code
CONDITIONALLY_ASSURED	Controlled code
LIMITED	Controlled code
ADVERSE_NOT_ASSURED	Controlled code
UNKNOWN_INCONCLUSIVE	Controlled code
SUSPENDED	Controlled code
WITHDRAWN	Controlled code
SUPERSEDED	Controlled code

Controlled vocabulary: OperatingState

Normative requirements.

Element	Requirement
NORMAL	Controlled code
MAINTENANCE	Controlled code
DEGRADED	Controlled code
EMERGENCY	Controlled code
RECOVERY	Controlled code
TRANSITION	Controlled code
UNKNOWN	Controlled code

Controlled vocabulary: Classification

Normative requirements.

Element	Requirement
PUBLIC	Controlled code
INTERNAL	Controlled code
CLIENT_RESTRICTED	Controlled code
CONFIDENTIAL	Controlled code
HIGHLY_RESTRICTED	Controlled code

12 Serialization architecture

Normative requirements.

Element	Requirement
JSON	Application exchange
JSON Schema 2020-12	Structural validation
JSON-LD 1.1	Linked data
RDF/Turtle	Graph exchange
OWL 2	Formal semantics
SHACL	Closed-world validation
CSV	Registries
NDJSON	Streams
Parquet	Non-authoritative analytics

13 JSON design rules

Normative requirements.

Element	Requirement
UTF-8	Required
camelCase	Required
ISO8601	Required
explicit-nullReason	Required
enum-codes	Required
stable-schema	Required
additionalProperties-false	Required
namespaced-extensions	Required
discriminator	Required
stable-references	Required
integrity-metadata	Required
classification	Required

14 Semantic graph rules

Normative requirements.

Element	Requirement
Protected JSON-LD terms	Required
Stable prefixes and typed IDs	Required
Separate AIDCF/AITG contexts	Required
OWL classes mirror registry	Required
No automatic approval through inference	Required
SHACL for closed-world cardinality	Required
Pinned imports	Required
Resolvable deprecations	Required

15 JSON Schema package

Normative requirements.

Element	Requirement
release-manifest.schema.json	Normative schema
common-envelope.schema.json	Normative schema
capability.schema.json	Normative schema
architecture.schema.json	Normative schema
dependency.schema.json	Normative schema
failure-scenario.schema.json	Normative schema
safeguard.schema.json	Normative schema
control.schema.json	Normative schema
evidence-requirement.schema.json	Normative schema
evidence-object.schema.json	Normative schema
assessment.schema.json	Normative schema
result-finding-unknown.schema.json	Normative schema
gate.schema.json	Normative schema
score.schema.json	Normative schema
decision.schema.json	Normative schema
program.schema.json	Normative schema
report-statement.schema.json	Normative schema
action-trigger-event.schema.json	Normative schema
integration-handoff.schema.json	Normative schema
validation-result.schema.json	Normative schema

16 Semantic publication package

Normative requirements.

Element	Requirement
aidcf-context-v1.0.jsonld	Context
aidcf-core-v1.0.owl	Ontology
aidcf-core-v1.0.ttl	Turtle
aidcf-shapes-v1.0.ttl	SHACL
registries/*.csv	Registries
examples/*.jsonld	Examples
conformance-tests/	Fixtures

17 Registry catalogue

Normative requirements.

Element	Requirement
artifacts.csv	code label definition owner version status source replacement
object-types.csv	code label definition owner version status source replacement
relationship-types.csv	code label definition owner version status source replacement
states.csv	code label definition owner version status source replacement
domains.csv	code label definition owner version status source replacement
control-families.csv	code label definition owner version status source replacement
controls.csv	code label definition owner version status source replacement
critical-gates.csv	code label definition owner version status source replacement
evidence-classes.csv	code label definition owner version status source replacement
evidence-quality.csv	code label definition owner version status source replacement
assessment-methods.csv	code label definition owner version status source replacement
finding-classes.csv	code label definition owner version status source replacement
unknown-reasons.csv	code label definition owner version status source replacement
decision-states.csv	code label definition owner version status source replacement
report-types.csv	code label definition owner version status source replacement
architecture-patterns.csv	code label definition owner version status source replacement
external-mappings.csv	code label definition owner version status source replacement
deprecated-terms.csv	code label definition owner version status source replacement
migration-rules.csv	code label definition owner version status source replacement

18 Business rules

Normative requirements.

Element	Requirement
BR-001	Applied control requires subject/applicability
BR-002	N/A requires rationale; missing evidence is UNKNOWN
BR-003	Positive result cannot exceed assessed dimension
BR-004	Gate PASS requires evidence
BR-005	Reasonably Assured blocked by gate FAIL/critical UNKNOWN
BR-006	Scores require coverage/confidence/profile/gates
BR-007	Report cannot exceed decision
BR-008	Closure requires evidence/reassessment
BR-009	Supersession preserves history
BR-010	AITG mapping preserves ownership
BR-011	Inference cannot auto-promote
BR-012	Sensitive export requires authority
BR-013	Material change triggers impact review
BR-014	Risk acceptance is not conformity
BR-015	Evidence reuse requires assertion review

19 Canonical queries

Normative requirements.

Element	Requirement
Q01	Capability to systems/dependencies
Q02	Outcome to failure paths/fault domains
Q03	Control to evidence/results/findings
Q04	Evidence to assertions/lineage/decisions
Q05	Gate to criterion/evidence/UNKNOWN/effect
Q06	Decision to gates/UNKNOWN/scores/approvals
Q07	Statement to decision consistency
Q08	Event to impacted objects
Q09	Provider to concentration/common modes
Q10	Snapshot to successor/migration
Q11	Candidates awaiting review
Q12	Sensitive paths by disclosure profile

20 Event model

Normative requirements.

Element	Requirement
eventId	Required
eventType	Required
source	Required
subjectRef	Required
observedAt	Required
effectiveAt	Required
payload	Required
schemaVersion	Required
sourceHealth	Required
severity	Required
materiality	Required
affectedAssertions	Required
candidateTriggers	Required
humanDisposition	Required
resultingDecision	Required
snapshot	Required
supersession	Required

21 Event types

Normative requirements.

Element	Requirement
ARCHITECTURE_CHANGED	Candidate event; human review required
CAPACITY_THRESHOLD_BREACHED	Candidate event; human review required
CONFIGURATION_CHANGED	Candidate event; human review required
PROVIDER_CHANGED	Candidate event; human review required
SOURCE_UNAVAILABLE	Candidate event; human review required
EVIDENCE_EXPIRED	Candidate event; human review required
EVIDENCE_CONFLICT	Candidate event; human review required
INCIDENT_DETECTED	Candidate event; human review required
TEST_FAILED	Candidate event; human review required
GATE_STATUS_CHANGED	Candidate event; human review required
FINDING_CLOSED	Candidate event; human review required
CONTROL_DRIFT	Candidate event; human review required
DEPENDENCY_DRIFT	Candidate event; human review required
DECISION_SUSPENDED	Candidate event; human review required
REPORT_WITHDRAWN	Candidate event; human review required

22 Change-impact contract

Normative requirements.

Element	Requirement
1	Validate event
2	Resolve identity/snapshot
3	Traverse assertions/dependencies
4	Identify controls/scenarios/gates
5	Recompute candidate coverage/scores
6	Apply candidate ceilings
7	Route review
8	Approve disposition
9	Create superseding snapshot
10	Notify report owners

23 API and exchange

Normative requirements.

Element	Requirement
versioned-media-type	Required
idempotency	Required
optimistic-concurrency	Required
immutable-snapshots	Required
pagination	Required
filtering	Required
bulk-validation	Required
dry-run	Required
explicit-errors	Required
classification-aware	Required
audit-log	Required
no-destructive-overwrite	Required

24 Semantic operations

Normative requirements.

Element	Requirement
validateRecord	Expected operation
validatePackage	Expected operation
resolveIdentifier	Expected operation
getLineage	Expected operation
getTraceabilityPath	Expected operation
getImpactAnalysis	Expected operation
createCandidateRecord	Expected operation
submitForReview	Expected operation
approveRecord	Expected operation
supersedeRecord	Expected operation
publishSnapshot	Expected operation
exportDisclosurePackage	Expected operation
ingestEvent	Expected operation
getConformanceResult	Expected operation

25 Security/privacy/disclosure

Normative requirements.

Element	Requirement
classification	Required
attribute-access	Required
edge-access	Required
purpose-bound-disclosure	Required
redaction	Required
encryption	Required
integrity	Required
no-secrets	Required
query-monitoring	Required
export-controls	Required
segregation	Required
minimisation	Required
retention	Required
breach-impact	Required
AI-derived-candidate	Required

26 Provenance/integrity

Normative requirements.

Element	Requirement
sourceSystem	Required
sourceRecord	Required
creator	Required
collector	Required
method	Required
timestamps	Required
clock	Required
transformationVersion	Required
hash	Required
signature	Required
custodian	Required
classification	Required
chainOfCustody	Required
review	Required
disposition	Required
supportLinks	Required
challengeLinks	Required
supersession	Required
migration	Required

27 Versioning/migration

Normative requirements.

Element	Requirement
Semantic versioning	Rule
Major for breaking	Rule
Minor for additive optional	Rule
Patch for metadata	Rule
Source/target migration	Rule
No automatic reinterpretation	Rule
Deprecated terms resolvable	Rule
Invalid migration blocked/UNKNOWN	Rule
Migration and validation records	Rule
Decision/report revalidation	Rule

28 Source-artifact binding

Normative requirements.

Element	Requirement
003	Ontology/taxonomy
004	Failure model
005	AITG integration
006	Principles
007	Controls
008	Evidence
009	Assessment
010	Scoring/decision
011	Program
012	Reporting
013	Architecture

29 Exchange bundles

Normative requirements.

Element	Requirement
Baseline	Manifest, records, validation and classification
Architecture	Manifest, records, validation and classification
Scenario	Manifest, records, validation and classification
Control	Manifest, records, validation and classification
Evidence	Manifest, records, validation and classification
Assessment	Manifest, records, validation and classification
Decision	Manifest, records, validation and classification
Reporting	Manifest, records, validation and classification
Continuing	Manifest, records, validation and classification
Integration	Manifest, records, validation and classification

30 Synthetic dataset

Normative requirements.

Element	Requirement
Capability/outcomes	Required synthetic fixture
Training chain	Required synthetic fixture
Power/cooling/compute/storage/network	Required synthetic fixture
External dependencies	Required synthetic fixture
Fault domains/common modes	Required synthetic fixture
Failure scenarios	Required synthetic fixture
Safeguards	Required synthetic fixture
Controls/evidence	Required synthetic fixture
At least one UNKNOWN	Required synthetic fixture
Gates	Required synthetic fixture
Score/decision	Required synthetic fixture
Reports	Required synthetic fixture
Change/reassessment	Required synthetic fixture
AITG handoff	Required synthetic fixture

31 Conformance tests

Normative requirements.

Element	Requirement
CT-01	Positive records
CT-02	Missing mandatory
CT-03	UNKNOWN handling
CT-04	Gate dominance
CT-05	Score reproducibility
CT-06	Report consistency
CT-07	Supersession
CT-08	Migration
CT-09	Security
CT-10	AITG handoff
CT-11	Event impact
CT-12	Anti-gaming
CT-13	Baseline conflict
CT-14	Performance

Template: Release Manifest

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

Template: Baseline Issue

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

Template: Object Registry Entry

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

Template: Relationship Entry

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

Template: Traceability Link

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

Template: Validation Result

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

Template: Migration Record

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

Template: Impact Analysis

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

Template: Disclosure Profile

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

Template: Conformance Report

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

Template: Integration Handoff

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

Template: Source-System Mapping

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

Template: Extension Registration

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

Template: Schema Change Request

Normative requirements.

Element	Requirement
id	
type	
version	
status	
scope	
sources	
subjects	
rule	
evidence	
lineage	
owner	
reviewer	
approval	
effectivePeriod	
classification	
supersession	

32 Data governance roles

Normative requirements.

Element	Requirement
Methodology authority	Normative meanings/release
Registry steward	Codes/mappings
Schema maintainer	Schemas/context/ontology/shapes
Domain steward	Domain mappings
Evidence steward	Evidence semantics
Decision steward	Gate/score/decision
Security steward	Classification/disclosure
AITG steward	Identity/handoff
Tool implementer	Implementation/audit
Independent reviewer	Conformance challenge
Release authority	Publication/migration

33 Extension governance

Normative requirements.

Element	Requirement
namespace	Required
owner	Required
justification	Required
targets	Required
no-collision	Required
datatype	Required
cardinality	Required
classification	Required
schema	Required
shape	Required
examples	Required
tests	Required
approval	Required
deprecation	Required
migration	Required

34 Data quality

Normative requirements.

Element	Requirement
Completeness	Measured/reported separately from assurance confidence
Validity	Measured/reported separately from assurance confidence
Consistency	Measured/reported separately from assurance confidence
Uniqueness	Measured/reported separately from assurance confidence
Timeliness	Measured/reported separately from assurance confidence
Accuracy	Measured/reported separately from assurance confidence
Lineage	Measured/reported separately from assurance confidence
Integrity	Measured/reported separately from assurance confidence
Reviewability	Measured/reported separately from assurance confidence
Security	Measured/reported separately from assurance confidence

35 Reference implementation

Normative requirements.

Element	Requirement
ArtifactRepository	Technology-neutral logical component
RegistryRepository	Technology-neutral logical component
SchemaRegistry	Technology-neutral logical component
IdentifierService	Technology-neutral logical component
IngestionService	Technology-neutral logical component
ValidationEngine	Technology-neutral logical component
TraceabilityGraph	Technology-neutral logical component
EvidenceConnectors	Technology-neutral logical component
AssessmentWorkflow	Technology-neutral logical component
GateScoreEngine	Technology-neutral logical component
DecisionReviewWorkflow	Technology-neutral logical component
ReportingService	Technology-neutral logical component
EventImpactService	Technology-neutral logical component
DisclosureService	Technology-neutral logical component
AuditLog	Technology-neutral logical component
ConformanceRunner	Technology-neutral logical component

36 Storage projections

Normative requirements.

Element	Requirement
Canonical object retained	Rule
Graph edges retain provenance/version	Rule
RDF named snapshots/shapes	Rule
Relational stable keys	Rule
Search non-authoritative	Rule
Warehouse snapshot dimensions	Rule
Evidence blobs referenced	Rule
Caches cannot approve/overwrite	Rule

Worked trace: Electrical

Normative requirements.

Element	Requirement
Trace	Outcome → power → dependencies → failure → safeguards → controls/evidence → CG-05 → decision
Forward	Outcome to report
Reverse	Claim to evidence/source
UNKNOWN	Missing link explicit
Change	Affected objects identified
Review	Candidate inference remains unapproved

Worked trace: Thermal

Normative requirements.

Element	Requirement
Trace	Outcome → cooling → common mode → scenario → controls/evidence → CG-06/09/10
Forward	Outcome to report
Reverse	Claim to evidence/source
UNKNOWN	Missing link explicit
Change	Affected objects identified
Review	Candidate inference remains unapproved

Worked trace: Digital authority

Normative requirements.

Element	Requirement
Trace	Outcome → identity/controller → safeguards → CG-07/15
Forward	Outcome to report
Reverse	Claim to evidence/source
UNKNOWN	Missing link explicit
Change	Affected objects identified
Review	Candidate inference remains unapproved

Worked trace: Recovery

Normative requirements.

Element	Requirement
Trace	Outcome → restore/alternate capacity → dependencies → exercise → CG-03/08/09/10
Forward	Outcome to report
Reverse	Claim to evidence/source
UNKNOWN	Missing link explicit
Change	Affected objects identified
Review	Candidate inference remains unapproved

Worked trace: Report lineage

Normative requirements.

Element	Requirement
Trace	Decision → gates/UNKNOWN/scores → statement → validity → trigger → supersession
Forward	Outcome to report
Reverse	Claim to evidence/source
UNKNOWN	Missing link explicit
Change	Affected objects identified
Review	Candidate inference remains unapproved

37 Conformance validation

Normative requirements.

Element	Requirement
TDS-VAL-001	Manifest approved
TDS-VAL-002	Baseline conflicts visible
TDS-VAL-003	IDs stable/unique
TDS-VAL-004	Owner/source/version/status present
TDS-VAL-005	Relationships or UNKNOWN
TDS-VAL-006	N/A justified
TDS-VAL-007	Assertions evidenced/UNKNOWN
TDS-VAL-008	Evidence quality claim-specific
TDS-VAL-009	Results link method/subject/evidence/reviewer
TDS-VAL-010	Gate PASS has evidence
TDS-VAL-011	Scores reproducible
TDS-VAL-012	Decision respects gates/UNKNOWN
TDS-VAL-013	Report matches decision
TDS-VAL-014	Events do not auto-approve
TDS-VAL-015	History preserved
TDS-VAL-016	Migrations explicit
TDS-VAL-017	AITG ownership preserved
TDS-VAL-018	Sensitive paths controlled
TDS-VAL-019	Extensions non-colliding
TDS-VAL-020	Tests pass

38 Review and release gates

Normative requirements.

Element	Requirement
Methodology author	Pending
Ontology/data architect	Pending
Semantic reviewers	Pending
Control/evidence reviewer	Pending
Assessment reviewer	Pending
Scoring/decision reviewer	Pending
Program/reporting reviewer	Pending
Reference-architecture reviewer	Pending
JSON Schema reviewer	Pending
RDF/OWL/SHACL reviewer	Pending
Security/privacy reviewer	Pending
AITG reviewer	Pending
Synthetic dataset pilot	Pending
Graph pilot	Pending
API/event pilot	Pending
Independent conformance review	Pending
Public release	Pending

Final doctrine

AIDCF-014 makes assurance claims reconstructable and testable without giving machines authority they do not possess. Structural validity does not prove truth, engineering adequacy or assurance. Conflict, incomplete lineage and missing evidence remain explicit and prevent stronger conclusions.